Project Report: HR Retention Analysis and Prediction

Course: ASDS 5302 Team: Group 5 (Mavericks Data)

Introduction

Employee retention is a critical metric for organizational stability and growth. This project, titled "HR Data Employment Retention Prediction," utilizes a comprehensive human resources dataset to analyze patterns in workforce attrition. By integrating data analytics, machine learning, and visual storytelling via Tableau, our team aims to transform raw HR data into actionable business insights.

Problem Statement

The primary research question driving this analysis is: **Can employee retention be analyzed and predicted using a company's HR data?**. To answer this, the project focuses on two distinct but complementary objectives:

1. **Prediction:** Building a machine learning model to identify which employees are at risk of termination.
2. **Diagnostics:** Using visual analytics to understand the structural and environmental reasons *why* employees are leaving.

Dataset Scope

The analysis is based on a dataset containing **8,950 employee records**. The data consists of **15 variables**, covering key employee attributes such as demographics, job roles, compensation, and performance ratings. The target variable for this study is binary classification based on the termdate (termination date) field, distinguishing between "Active" and "Terminated" employees.

Data Preprocessing and Descriptive Analysis

The goal of this stage was to prepare the Human Resources dataset for HR retention analysis before building any predictive models. Data preprocessing focused on data quality, missingness, and creating new variables for analysis. The descriptive analysis then examined how employee characteristics are distributed across the dataset and how these relate to termination status at an observational level.

Import Data

The first part of this project focused on preparing the HR dataset so that the analysis would be accurate and reliable. The dataset was imported into a pandas DataFrame using a CSV file. Immediately after loading, functions such as df.info(), df.head(), df.shape, and df.dtypes were used to understand the basic structure of the data. These checks confirmed the number of rows and columns (employee-level observations and variables), which columns were stored as numeric or categorical, and whether any variables looked suspicious in terms of type.

The first several rows showed a reasonable HR structure including employee identifiers, demographic fields (gender, birthdate, and state), job-related fields (department, job title, overtime status, performance rating, education level), and compensation and date fields (salary, hire date, and termination date).

Detect Duplicate Rows and Cleaning

To ensure each employee was only represented once, duplicate rows were checked using *dup_count = df.duplicated().sum()*. When duplicates were found, they were removed with *df = df.drop_duplicates()*, and the dataset shape was reinspected. This step ensured that subsequent frequency counts, proportions, and models were not distorted by repeated employee records.

Check Initial Missing Value

Missing values in the dataset were identified using df.isna().sum(). A large number of missing values of 5,288 were observed in the termdate column. This step only inspected missing values without modifying the data. The goal was to determine which variables had incomplete information before making decisions about imputation or transformation.

Imputation and Reverting to the Original Data

To explore how missing values could be addressed, a temporary imputation step was implemented to see how a generic imputation approach would behave.

Numeric columns were identified with *num_cols = df.select_dtypes(include=[np.number])*, and categorical columns with *cat_cols = df.select_dtypes(include="object")*.

The ***median*** was used for imputing numeric variables because it is a robust measure of central tendency that is not overly influenced by extreme values.

The ***mode*** was used for categorical variables because it keeps the original category structure and prevents creating values that do not exist in the dataset.

After testing this approach and rechecking missing values, the dataset was reloaded from the original csv file instead of applying the imputed version. This decision was made because the primary source of the missing variable was the termdate variable, and this reflects not terminated employment status rather than true data loss. Since active employees do not yet have termination dates, imputing values would misrepresent the meaning of the terminated variable. Therefore, the original dataset was restored for HR retention analysis.

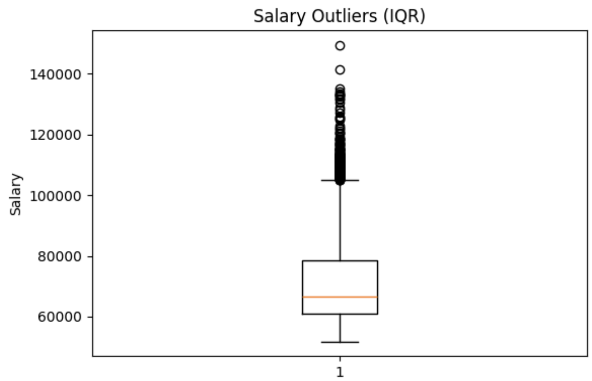
Create the “Terminated” Variable

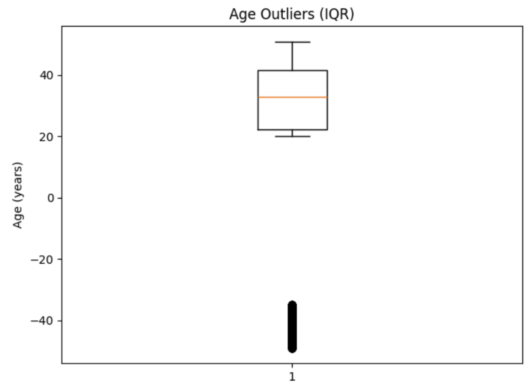
A new binary variable named terminated was created based on the termdate column. If termdate was non-missing, the employee was labeled as 1 (terminated). If the term date was missing, the employee was labeled as 0 (not terminated). The distribution of this variable verifies that both terminated and non-terminated employees were properly represented in the dataset. This variable is the main variable we used throughout the descriptive analyses and modeling.

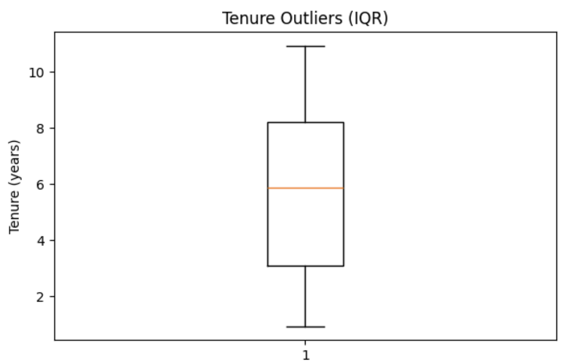
Outliers & Boxplots

The Interquartile Range (IQR) method was applied to the salary, age_years, and tenure_years variables to detect potential numerical outliers. For each variable, Q1, Q3, IQR, and corresponding lower and upper bounds were computed. Values outside these bounds were flagged and counted.

Boxplots were created for salary, age, and tenure to visualize their distributions and highlight any extreme values. The salary and age plots showed the presence of high-end outliers, while tenure did not show extreme values beyond the IQR limits. After reviewing these distributions, the outliers were retained because they are plausible in an HR context, such as highly paid staff or older workers, and do not appear as data errors. Therefore, removing them would have eliminated the real workforce variance and weakened the validity of the HR retention analysis.



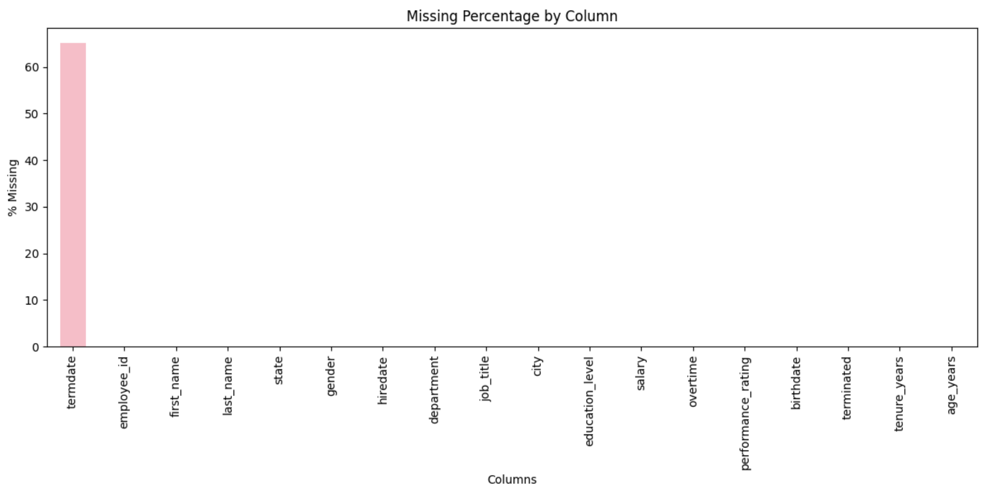


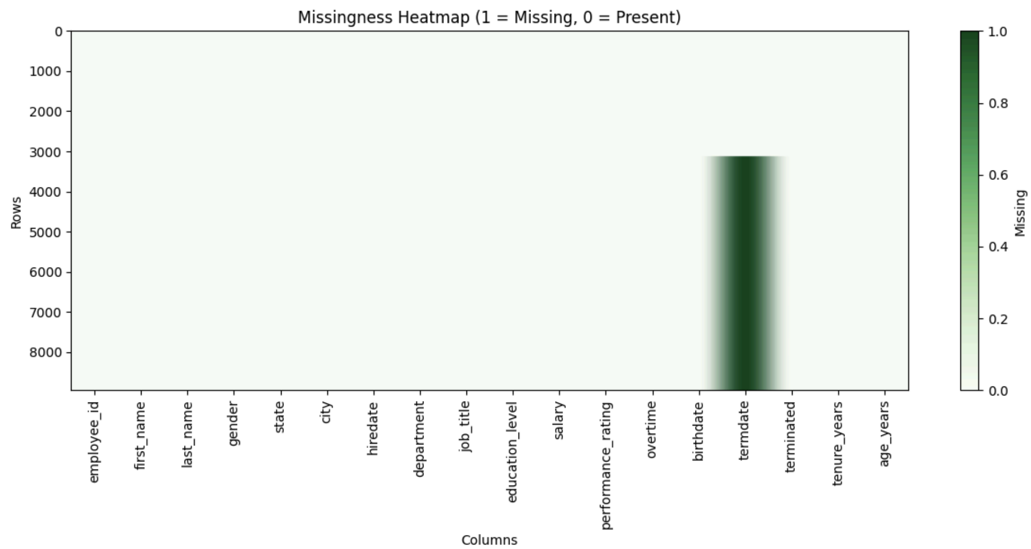


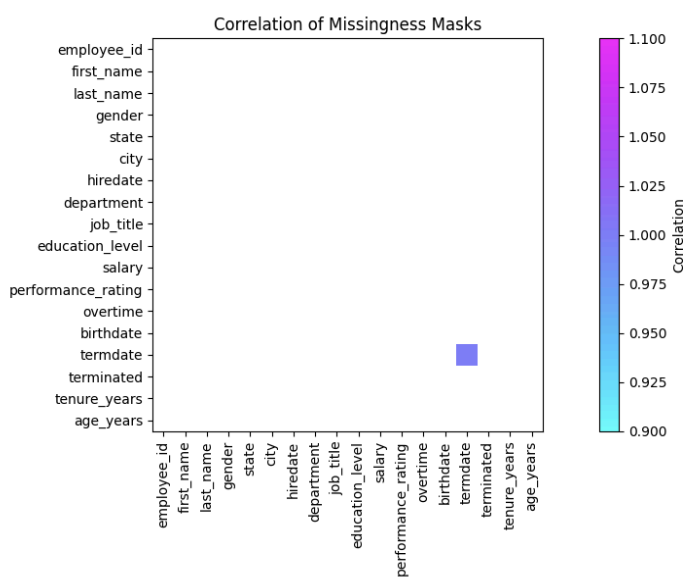
Missing Data Analysis: Missing % by column, Heatmap of missingness patterns, Correlation of missingness masks

The bar chart of missing % by column shows that the termdate variable is the only column with more than 60 percent missing values. This result is expected because most employees in the dataset are still actively employed and therefore do not yet have a termination date. A heatmap was created to visualize the overall pattern of missing values across the dataset. The heatmap displays a strong vertical band of missing values only in the termdate column, confirming that missingness is concentrated in a single variable rather than spread across multiple columns.

In addition, a correlation analysis of missingness masks shows no meaningful relationship between termdate and other variables. Overall, these results indicate that the dataset is largely complete and that the missing values are driven by employment status rather than data quality issues.


Missing Percentage by Column
% Missing
Columns
0
10
20
30
40
50
60
termdate
employee_id
first_name
last_name
state
gender
hiredate
department
job_title
city
education_level
salary
overtime
performance_rating
birthdate
terminated
tenure_years
age_years


Missingness Heatmap (1 = Missing, 0 = Present)
Rows
Columns
Missing
0
1000
2000
3000
4000
5000
6000
7000
8000
0.0
0.2
0.4
0.6
0.8
1.0
employee_id
first_name
last_name
gender
state
city
hiredate
department
job_title
education_level
salary
performance_rating
overtime
birthdate
termdate
terminated
tenure_years
age_years


Correlation of Missingness Masks
Correlation
0.900
0.925
0.950
0.975
1.000
1.025
1.050
1.075
1.100
employee_id
first_name
last_name
gender
state
city
hiredate
department
job_title
education_level
salary
performance_rating
overtime
birthdate
termdate
terminated
tenure_years
age_years

Employee Counts and Gender Distribution

This step showed the overall structure of the workforce by looking at how employees are distributed across key categories such as department, education level, overtime status, performance rating, job title, birthdate, and hire date. These counts provided a clear picture of how the workforce is organized before looking at termination patterns or building predictive models. The employee counts by department also showed how workers are spread across different areas of the company and helped identify which departments are larger or smaller. After generating these overall category counts, gender based tables were created to show how male and female employees are distributed across different organizational and demographic groups. This provided important background information for later HR retention analysis.

Employee Counts and Terminated Variable

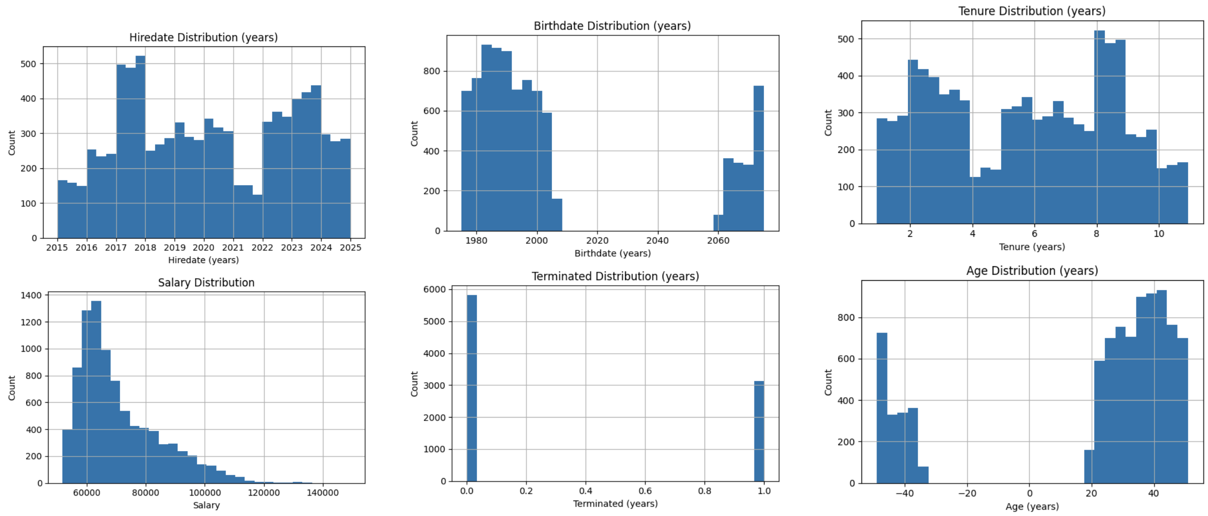
This step showed how employee termination varies in several workforces by gender, education level, department, overtime status, performance rating, job title, birthdate, and hire date. For each group, the mean, median, and count of the terminated variable were calculated to summarize termination patterns. Since the terminated variable is binary, the mean represents the average termination rate for each group, while the median confirms whether most employees in the group are terminated or not terminated. The count shows how many employees are included in each group so that termination patterns can be interpreted in the proper context. Grouping by gender gives a comparison of termination between male and female employees, while education level and department groupings show how termination differs by educational backgrounds and organizational units. The overtime, performance rating, and job title groupings provide insight into how work conditions, job performance, and role type relate to termination patterns. Overall, this step provides a clear descriptive summary of how termination is distributed across demographic, job-related, and time-based characteristics.

Summary Statistics for Numerical Variables Columns

All numeric variables, including hiredate, salary, birthdate, terminated, tenure_years, and age_years, contain 8,950 observations, with no missing values shown in the table. The average salary is about 70,946 with values ranging from 51,835 to 149,377, indicating a wide range in employee compensation. The mean tenure is approximately 5.72 years, and the terminated variable has a mean of 0.349, meaning that about 35% of employees in the dataset have been terminated. However, the birthdate and age_years variables show data issues, including a negative minimum age and a future maximum birthdate, which indicates errors in the date calculations that must be corrected in further analysis.

Histograms for Numerical Variables Columns

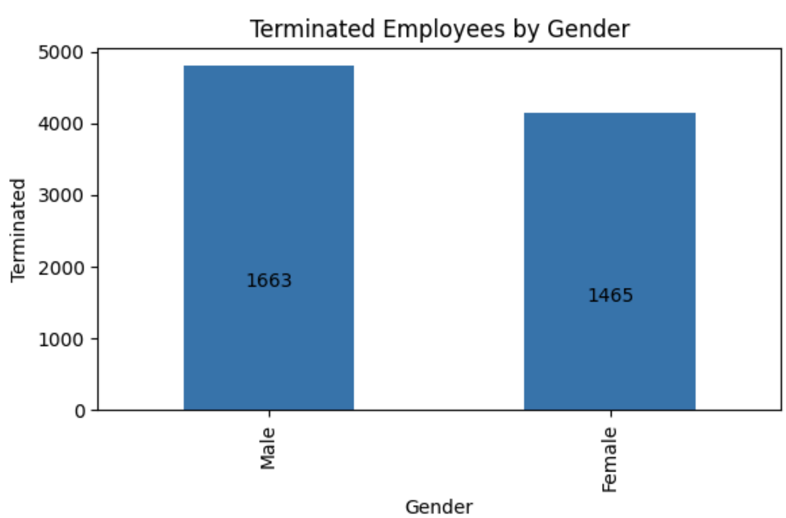
These histograms show how employees are distributed in numerical variables: hiredate, salary, birthdate, terminated, tenure_years, and age_years. The graph visually summarizes when employees were hired, how salaries are distributed, the range of birthdate distribution in years, the balance between terminated and not terminated employees, and how long employees typically remain with the company. The age and birthdate histograms are especially helpful for checking whether the age values are realistic and for identifying any remaining data errors.



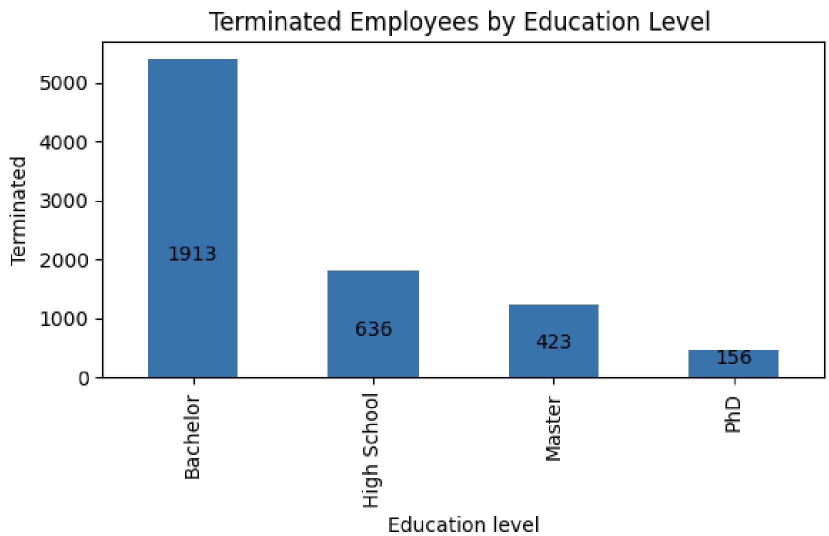
Bar Charts: Terminated Employees

The bar graphs show how terminated employees are distributed across key workforce categories, including gender, education level, department, overtime status, performance rating, job title, birthdate, and hiredate.

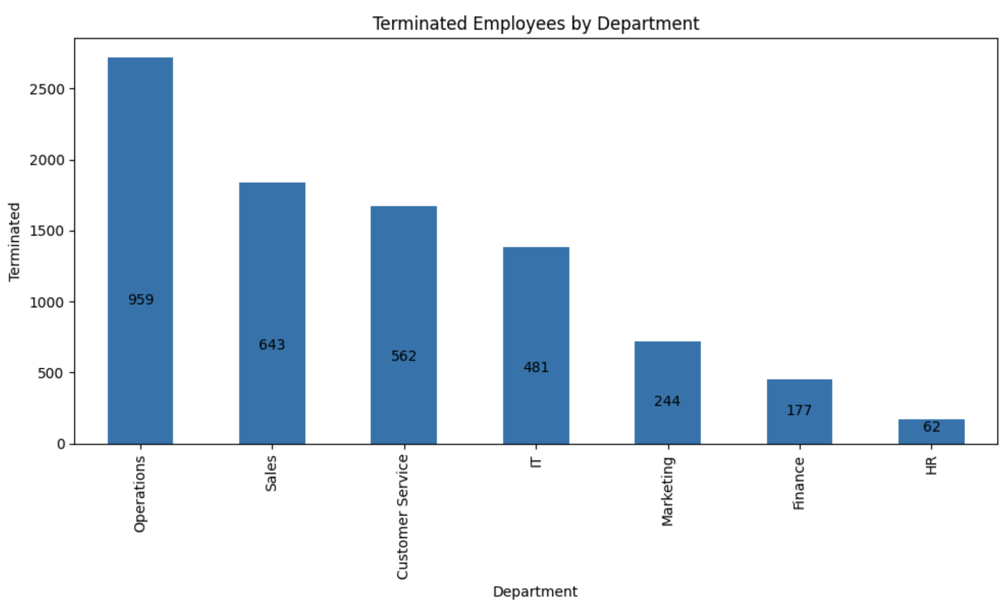
By gender, 1,663 males and 1,465 females were terminated. This shows a slightly higher number of terminations among male employees.



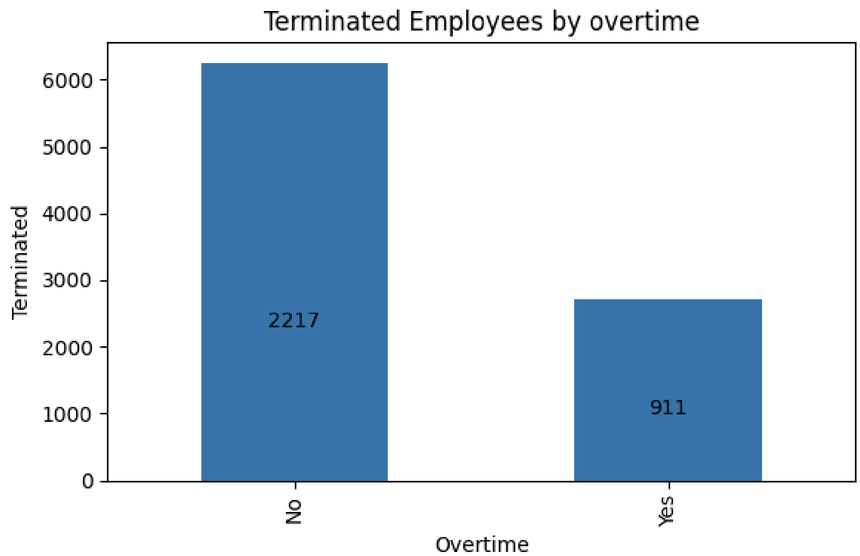
In terms of education, employees with a bachelor’s degree account for the highest number of terminations (1,913), followed by high school (636), master’s (423), and PhD (156).



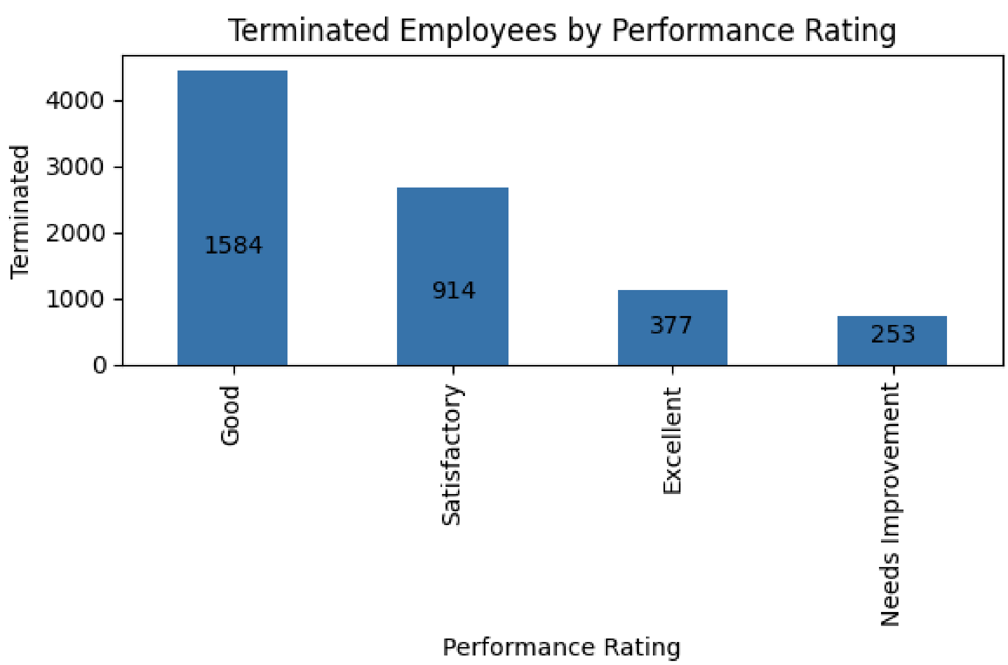
At the department, the most terminations occurred in Operations (959), followed by Sales (643), Customer Service (562), IT (481), Marketing (244), Finance (177), and HR (62).



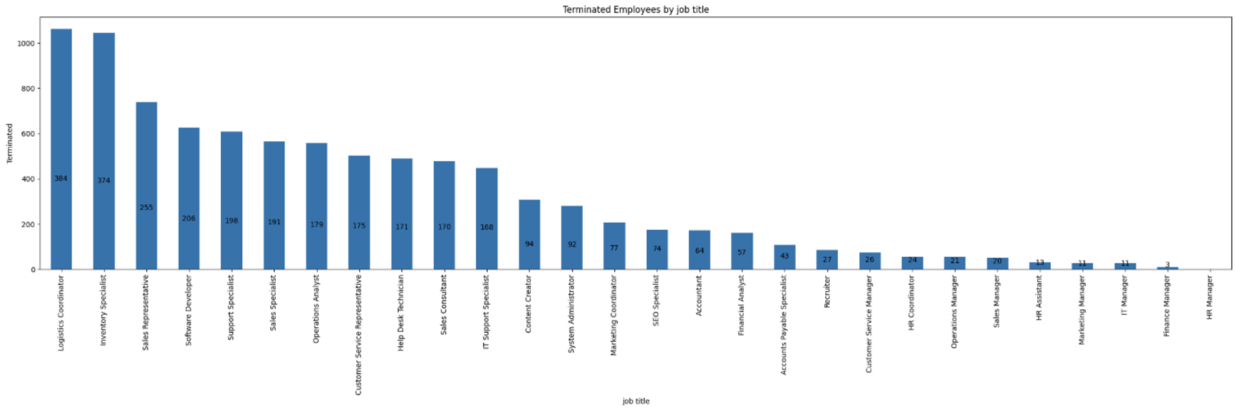
For overtime status, 2,217 terminated employees did not work overtime, compared to 911 who did, indicating that most terminations occurred among non-overtime workers.



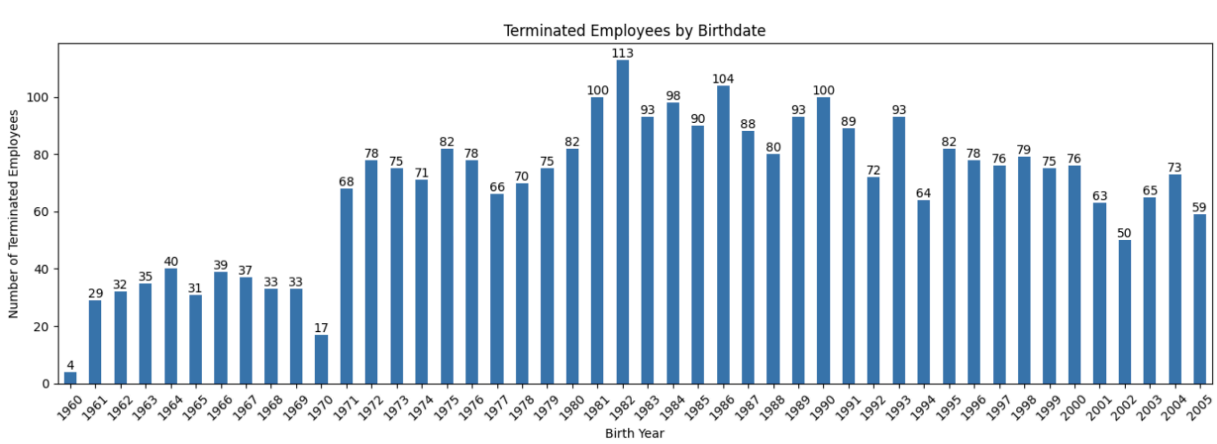
By performance rating, the highest number of terminations occurred among employees rated "Good" (1,584), followed by Satisfactory (914), Excellent (377), and Needs Improvement (253).



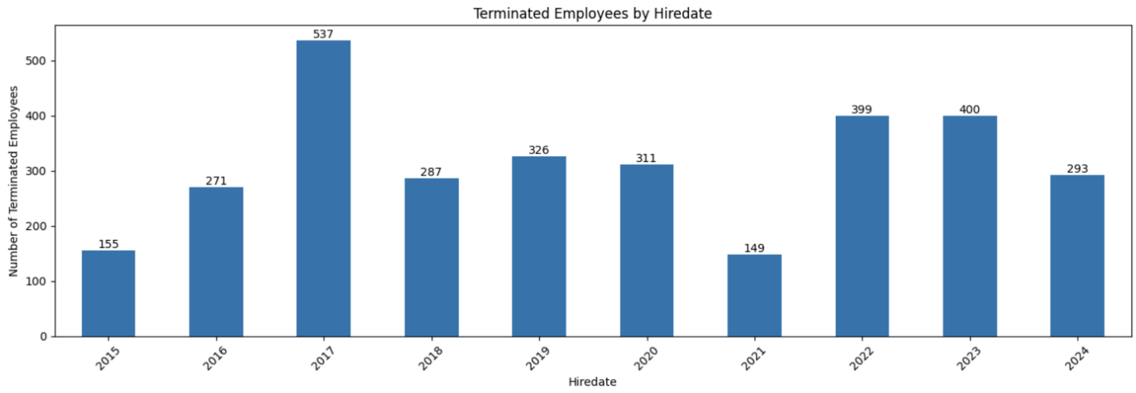
Across job titles, terminations are mainly concentrated in operational and support roles, especially Logistics Coordinator (384), Inventory Specialist (374), and Sales Representative (255), with gradually decreasing counts across technical, administrative, and managerial positions.



The birthdate bar chart shows that terminated employees are distributed across birth years from the 1960s through the early 2000s, with the highest counts clustered in the early to mid 1980s, especially 1981 (113) and 1985 (104). Terminations remain relatively steady through the late 1980s and 1990s, with lower counts at the youngest and oldest ends of the age range.



The hiredate bar chart shows that terminations occurred across all hiring years from 2015 to 2024, with the highest counts among employees hired in 2017 (537), followed by 2023 (400), 2022 (399), and 2019 (326), while fewer terminations occurred among employees hired in 2015 (155) and 2021 (149). This indicates that terminations are spread across multiple hiring periods rather than being driven by a single hiring cohort.



Lastly, the average salary of terminated employees (70,860) and not-terminated employees (70,992) is very similar. This indicates that termination is not strongly separated by salary at the descriptive level.

Conclusion: Data Preprocessing and Descriptive Analysis

In conclusion, the preprocessing and descriptive analysis confirm that the HR dataset is suitable for HR retention analysis. Data preprocessing focused on improving data quality by removing duplicate rows, checking for missing values, creating a new variable, and validating numerical variables such as age, tenure, and salary. The missing data analysis showed that the most missing values come from the termdate variable, which reflects active employment status rather than true data quality problems. A binary-terminated variable was created from termdate and used as the main outcome variable for all descriptive analyses. Outliers identified using the IQR method in salary and age were retained because they represent realistic workforce variation rather than data errors. The descriptive analysis showed how employee characteristics are distributed and how they relate to termination across gender, education level, department, overtime status, performance rating, job title, hire year, and birthdate. Overall, this dataset is

well structured and appropriate for HR retention modeling, although corrections to age and birthdate need to be addressed before modeling.
Modeling Approach and Results

Feature Engineering

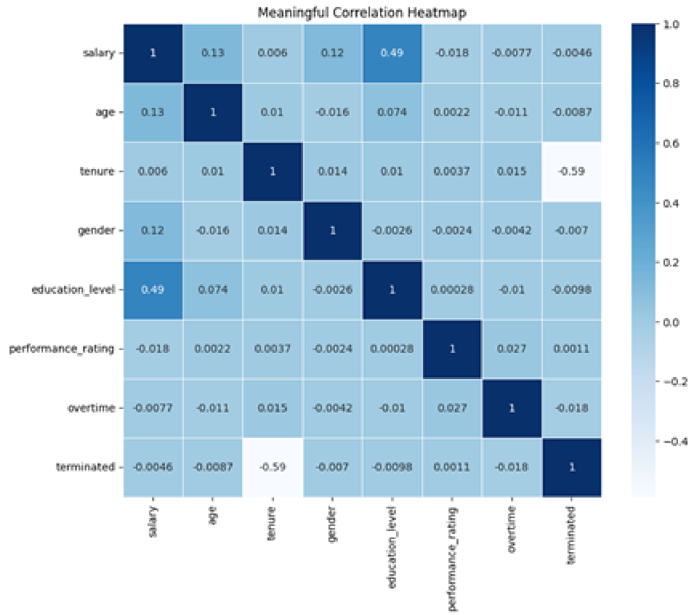
To prepare the dataset for predictive modeling, several new variables were created to better represent employee history and behavior. Age was calculated using birthdat,e and tenure was calculated using hiredate and termdate. These transformations allowed the model to capture patterns related to work length and employee experience. The overtime variable was converted into numeric form and the original date fields were removed after the necessary information was extracted. These steps improved data quality and created strong foundations for the later modeling stages.

Categorical columns were encoded to allow the model to process them effectively. Gender and overtime were encoded as binary values. Education level and performance rating were encoded using ordered values to reflect their natural progression. State, city, department, and job title were one-hot encoded to represent all categories without assigning artificial order.

	employee_id	first_name	last_name	gender	education_level	salary	performance_rating	overtime	terminated	age	tenure
0	00-95822412	Danielle	Johnson	0	1	81552	3	0	1	45	1
1	00-42868828	Jeffrey	Doyle	1	2	107520	3	0	1	37	3
2	00-83197857	Patricia	Miller	1	2	61104	3	1	1	31	0
3	00-13999315	Anthony	Robinson	1	2	73770	3	0	1	45	4
4	00-90801586	Anthony	Gonzalez	0	2	55581	2	0	1	40	6
5	00-97226012	Amy	Robinson	0	2	73800	3	1	1	35	1
6	00-70291817	Lisa	Smith	1	2	68181	3	0	1	32	0
7	00-31429110	Helen	Peterson	0	2	54499	3	1	1	28	0
8	00-30868105	Susan	Rogers	1	1	57340	3	0	1	23	4
9	00-22448136	Colin	Abbott	0	2	54065	3	1	1	44	4

Correlation Analysis

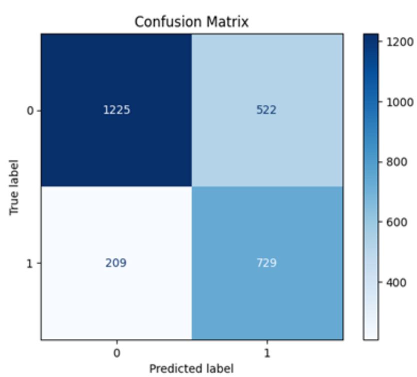
A focused correlation heatmap was created to explore relationships among the key numeric and ordinal variables. Tenure showed the strongest negative relationship with termination, which suggests that long-term employees are less likely to be terminated. Salary and education level showed a moderate positive relationship, while performance rating and overtime displayed minimal correlation with termination. These insights guided expectations for model behavior and feature influence.



Logistic Regression Modeling

The prepared dataset was used to train a logistic regression model. A stratified train test split ensured that the distribution of terminated and non-terminated employees remained consistent across both sets. Salary age and tenure were scaled using StandardScaler to support model stability. Class weight was set to balanced to account for unequal class sizes.

The initial model performed reasonably well and achieved strong recall for the terminated class. However, precision for this class was lower which indicated the presence of class imbalance.



	precision	recall	f1-score	support
0	0.85	0.70	0.77	1747
1	0.58	0.78	0.67	938
accuracy			0.73	2685
macro avg	0.72	0.74	0.72	2685
weighted avg	0.76	0.73	0.73	2685

Addressing Class Imbalance with SMOTE

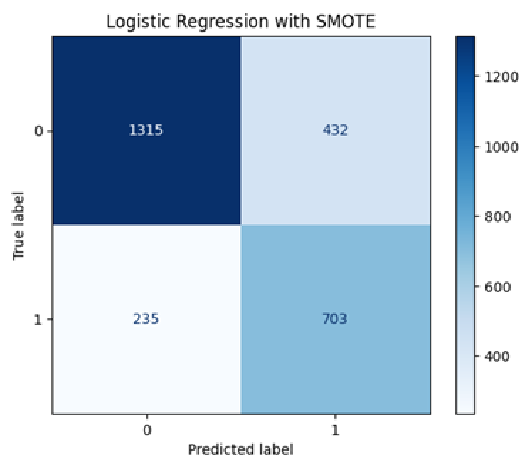
The terminated class made up a smaller portion of the dataset which affected model performance. To address this issue SMOTE was applied to the training set. SMOTE generates synthetic examples for the minority class and creates a balanced dataset without duplicating records. This allowed the model to learn termination patterns more effectively.

```
Before SMOTE
terminated
0    4075
1    2190
Name: count, dtype: int64

After SMOTE
terminated
0    4075
1    4075
Name: count, dtype: int64
```

Model Performance After SMOTE

After retraining the logistic regression model on the SMOTE balanced data performance improved for the terminated class. Recall and F1 score increased and the model showed stronger and more consistent behavior across both classes. These improvements confirmed that class balancing was an essential step for achieving fair and reliable predictions.




```
              precision    recall  f1-score   support

           0       0.85      0.75      0.80      1747
           1       0.62      0.75      0.68       938

    accuracy                           0.75      2685
   macro avg       0.73      0.75      0.74      2685
weighted avg       0.77      0.75      0.76      2685
```

Visual Analytics and Business Insights

While the Logistic Regression model successfully identified *who* is at risk of termination, we utilized Tableau to develop an interactive dashboard to understand *why* attrition is occurring. The dashboard serves as a diagnostic tool to contextualize the machine learning predictions and identify structural root causes within the organization.

Analysis of Demographic and Compensation Factors

A primary objective of the visual analysis was to validate or reject common assumptions regarding attrition, such as pay gaps or demographic bias.

- **Demographic Independence:** The visualization confirmed that termination is not driven by gender or age demographics. As noted in the analysis, these variables show zero meaningful correlation with termination rates.
- **Compensation Analysis:** Contrary to the hypothesis that employees are leaving for better pay, the dashboard revealed that salary distributions between active and terminated employees are nearly identical. The average salary for active employees is **$70,992**, compared to **$70,860** for terminated employees. This difference is statistically negligible, indicating that broad compensation adjustments are unlikely to solve the retention issue.

Identification of "Departmental Churn."

By filtering the data by department, the dashboard isolated the attrition problem to specific areas of the business rather than a company-wide culture issue. The churn is heavily concentrated in three high-volume departments:

1. **Operations**
2. **Sales**
3. **Customer Service**

These three units account for the vast majority of turnover, whereas departments like Finance and HR show significantly higher stability.

The "Mass Intake / Mass Exit" Cycle

A temporal analysis of Hire Date versus Term Date within the dashboard revealed a critical pattern in the problematic departments. These units are operating in a cycle of **"mass intake and mass exit"**—hiring in large volumes only to see equivalent termination spikes shortly thereafter.

- **The 2017 Anomaly:** This cycle peaked dramatically in **2017**, when **537 employees** hired that year were subsequently terminated. This is nearly double the rate of adjacent years (271 in 2016 and 287 in 2018), pointing to a specific operational or hiring failure during that fiscal period.

This pattern suggests that the root cause is structural within these specific units. Potential drivers identified include:

- Inadequate onboarding processes for high-volume roles.
- Role clarity issues leading to rapid burnout.
- Management quality or workload distribution within Operations and Sales teams.

Strategic Pivot

Based on the integration of the Logistic Regression model and Tableau insights, the organization must pivot its retention strategy. Since the data proves that pay and demographics are not the primary drivers, broad-spectrum solutions—such as company-wide pay raises or general diversity initiatives—will yield a low Return on Investment (ROI).

Targeted Interventions

We recommend a targeted approach focused on the "Departmental Churn" identified in the dashboard:

1. **Audit Operations and Sales:** Conduct a qualitative audit of the management practices and workload in the Operations, Sales, and Customer Service departments to identify why new hires churn so quickly.
2. **Revamp Onboarding:** Break the "mass intake/mass exit" cycle by restructuring the onboarding process for these specific high-volume roles to ensure better employee integration.
3. **Predictive Intervention:** Utilize the Logistic Regression model (trained with SMOTE) to flag high-risk employees specifically within these three departments for early intervention by HR.

Conclusion: By combining predictive modeling with visual analytics, we have determined that employee retention in this dataset is not a financial issue, but a structural one. The organization has the data required to predict *who* will leave, and now possesses the insights to fix the departmental structures causing them to leave.

Executive Summary

Executive Summary: This project explores employee retention using a dataset of 8,950 records. The team employed a multi-faceted approach: **Exploratory Data Analysis (EDA)** to clean and understand the data, **Machine Learning (Logistic Regression with SMOTE)** to predict termination risks, and **Tableau Dashboarding** to diagnose root causes.

Key findings include:

- **Modeling:** The application of SMOTE successfully balanced the dataset, allowing the Logistic Regression model to achieve higher recall and F1 scores for identifying terminated employees.
- **Insights:** Visual analysis debunked the theory that attrition is pay-driven (average salaries for active vs. terminated employees differ by less than $150). Instead, turnover is localized to the Operations, Sales, and Customer Service departments, which suffer from a cyclical "hire-burnout-terminate" pattern. The final report recommends targeted

structural interventions in these high-churn departments rather than broad organizational changes.

Click here to view the code

Click here to view the Dashboard